SPHERICAL BUSHINGS

Steel-on-steel Spherical Bushings Inch Series





Shaft dia. 12.700 — 63.500mm

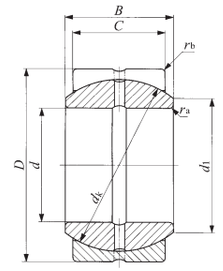
Shaft dia. mm (inch)	Identification number Without seal	With seals	Mass (Ref.) kg	Boundary dimensions mm(inch) d	D	B	C
12.700 (1/2)	**SBB 8**	—	0.020	12.700(1/2)	22.225(7/8)	11.10(.437)	9.52(.375)
15.875 (5/8)	**SBB 10**	—	0.036	15.875(5/8)	26.988(1 1/16)	13.89(.547)	11.91(.469)
19.050 (3/4)	**SBB 12**	**SBB 12-2RS**	0.057	19.050(3/4)	31.750(1 1/4)	16.66(.656)	14.27(.562)
22.225 (7/8)	**SBB 14**	**SBB 14-2RS**	0.088	22.225(7/8)	36.512(1 7/16)	19.43(.765)	16.66(.656)
25.400 (1)	**SBB 16**	**SBB 16-2RS**	0.125	25.400(1)	41.275(1 5/8)	22.22(.875)	19.05(.750)
31.750 (1 1/4)	**SBB 20**	**SBB 20-2RS**	0.234	31.750(1 1/4)	50.800(2)	27.76(1.093)	23.80(.937)
34.925 (1 3/8)	**SBB 22**	**SBB 22-2RS**	0.349	34.925(1 3/8)	55.562(2 3/16)	30.15(1.187)	26.19(1.031)
38.100 (1 1/2)	**SBB 24**	**SBB 24-2RS**	0.424	38.100(1 1/2)	61.912(2 7/16)	33.32(1.312)	28.58(1.125)
44.450 (1 3/4)	**SBB 28**	**SBB 28-2RS**	0.649	44.450(1 3/4)	71.438(2 13/16)	38.89(1.531)	33.32(1.312)
50.800 (2)	**SBB 32**	**SBB 32-2RS**	0.939	50.800(2)	80.962(3 3/16)	44.45(1.750)	38.10(1.500)
57.150 (2 1/4)	**SBB 36**	**SBB 36-2RS**	1.32	57.150(2 1/4)	90.488(3 9/16)	50.01(1.969)	42.85(1.687)
63.500 (2 1/2)	**SBB 40**	**SBB 40-2RS**	1.85	63.500(2 1/2)	100.012(3 15/16)	55.55(2.187)	47.62(1.875)

Note(1) Maximum allowable corner radius of the shaft or housing

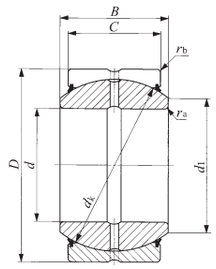
Remarks1. The value with mark * is applicable to types without seals. For types with seals, the value is 0.4 mm.

2. The inner ring and the outer ring have an oil groove and two oil holes, respectively.

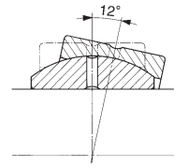
3. Not provided with prepacked grease. Perform proper lubrication for use.



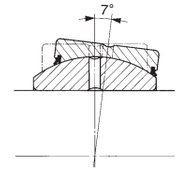
SBB



SBB····-2RS



SBB



SBB····-2RS

Permissible tilting angle

d_k	Radial internal clearance mm Min./Max.	Mounting dimensions mm d_1	$r_{as\,max}$ (1) Max.	$r_{bs\,max}$ (1) Max.	Dynamic load capacity C_d N	Static load capacity C_s N
18 (.709)	0.05／0.15	14.0	0.2	0.6	16 800	101 000
23 (.906)	0.05／0.15	17.9	0.2	0.8	26 900	161 000
27.5(1.083)	0.08／0.18	21.4	0.6	*0.8	38 500	231 000
32 (1.260)	0.08／0.18	25.0	0.6	*0.8	52 300	314 000
36 (1.417)	0.08／0.18	28.0	0.6	*0.8	67 300	404 000
45 (1.772)	0.08／0.18	35.1	0.6	0.8	105 000	630 000
49 (1.929)	0.08／0.18	38.5	0.6	0.8	126 000	755 000
55 (2.165)	0.08／0.18	43.3	0.6	0.8	154 000	925 000
64 (2.520)	0.08／0.18	50.4	0.6	0.8	209 000	1 250 000
73 (2.874)	0.08／0.18	57.6	0.6	0.8	273 000	1 640 000
82 (3.228)	0.10／0.20	64.9	0.6	0.8	345 000	2 070 000
91 (3.583)	0.10／0.20	72.0	0.6	0.8	425 000	2 550 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SPHERICAL BUSHINGS

Steel-on-steel Spherical Bushings Inch Series

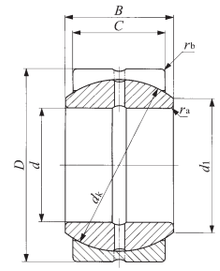




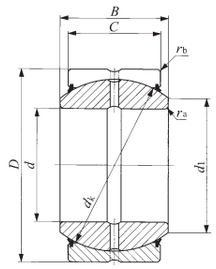
Shaft dia. 69.850 — 152.400mm

Shaft dia. mm (inch)	Identification number Without seal	With seals	Mass (Ref.) kg	Boundary dimensions mm(inch) d	D	B	C
69.850 (2¾)	SBB 44	SBB 44-2RS	2.44	69.850 (2¾)	111.125 (4⅜)	61.11 (2.406)	52.37 (2.062)
76.200 (3)	SBB 48	SBB 48-2RS	3.12	76.200 (3)	120.650 (4¾)	66.68 (2.625)	57.15 (2.250)
82.550 (3¼)	SBB 52	SBB 52-2RS	3.92	82.550 (3¼)	130.175 (5⅛)	72.24 (2.844)	61.90 (2.437)
88.900 (3½)	SBB 56	SBB 56-2RS	4.83	88.900 (3½)	139.700 (5½)	77.77 (3.062)	66.68 (2.625)
95.250 (3¾)	SBB 60	SBB 60-2RS	5.87	95.250 (3¾)	149.225 (5⅞)	83.34 (3.281)	71.42 (2.812)
101.600 (4)	SBB 64	SBB 64-2RS	7.07	101.600 (4)	158.750 (6¼)	88.90 (3.500)	76.20 (3.000)
107.950 (4¼)	SBB 68	SBB 68-2RS	8.46	107.950 (4¼)	168.275 (6⅝)	94.46 (3.719)	80.95 (3.187)
114.300 (4½)	SBB 72	SBB 72-2RS	9.94	114.300 (4½)	177.800 (7)	100.00 (3.937)	85.72 (3.375)
120.650 (4¾)	SBB 76	SBB 76-2RS	11.6	120.650 (4¾)	187.325 (7⅜)	105.56 (4.156)	90.47 (3.562)
127.000 (5)	SBB 80	SBB 80-2RS	13.5	127.000 (5)	196.850 (7¾)	111.12 (4.375)	95.25 (3.750)
152.400 (6)	SBB 96	SBB 96-2RS	17.6	152.400 (6)	222.250 (8¾)	120.65 (4.750)	104.78 (4.125)

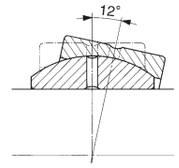
Note(1) Maximum allowable corner radius of the shaft or housing
Remarks1. The inner ring and the outer ring have an oil groove and two oil holes, respectively.
2. Not provided with prepacked grease. Perform proper lubrication for use.



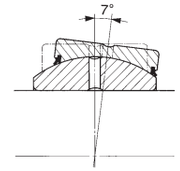
SBB



SBB····-2RS



SBB



SBB····-2RS

Permissible tilting angle

d_k	Radial internal clearance mm Min./Max.	Mounting dimensions mm d_1	$r_{as\,max}$ (1) Max.	$r_{bs\,max}$ (1) Max.	Dynamic load capacity C_d N	Static load capacity C_s N
100(3.937)	0.10／0.20	79.0	0.6	0.8	514 000	3 080 000
110(4.331)	0.10／0.20	86.5	0.6	0.8	616 000	3 700 000
119(4.685)	0.13／0.23	94.1	0.6	0.8	722 000	4 330 000
128(5.039)	0.13／0.23	101.6	0.6	0.8	837 000	5 020 000
137(5.394)	0.13／0.23	108.4	0.6	0.8	960 000	5 760 000
146(5.748)	0.13／0.23	115.8	0.6	0.8	1 090 000	6 550 000
155(6.102)	0.13／0.23	122.6	0.8	1.1	1 230 000	7 380 000
164(6.457)	0.13／0.23	129.8	0.8	1.1	1 380 000	8 270 000
173(6.811)	0.13／0.23	136.8	0.8	1.1	1 530 000	9 210 000
183(7.205)	0.13／0.23	144.9	0.8	1.1	1 710 000	10 300 000
207(8.150)	0.13／0.23	167.5	0.8	1.1	2 130 000	12 800 000

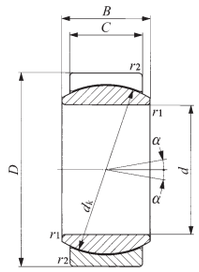
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

SPHERICAL BUSHINGS

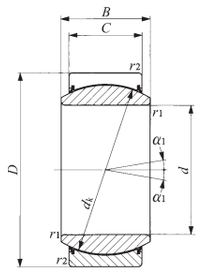
Maintenance-free Spherical Bushings



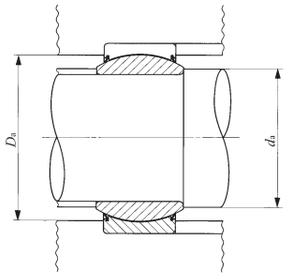




GE···EC



GE···EC-2RS



Shaft dia. 15 — 70mm

Shaft dia. mm	Identification number		Mass (Ref.)	Boundary dimensions mm							Permissible tilting angle degree	
	Without seals	With seals	kg	d	D	B	C	d_k	$r_{1s\,min}$ (1)	$r_{2s\,min}$ (1)	α	α_1
15	**GE 15EC**	—	0.032	15	26	12	9	22	0.3	0.3	8	—
17	**GE 17EC**	—	0.049	17	30	14	10	25	0.3	0.3	10	—
20	**GE 20EC**	—	0.065	20	35	16	12	29	0.3	0.3	9	—
25	**GE 25EC**	—	0.115	25	42	20	16	35.5	0.6	0.6	7	—
30	**GE 30EC**	**GE 30EC-2RS**	0.160	30	47	22	18	40.7	0.6	0.6	6	4
35	—	**GE 35EC-2RS**	0.258	35	55	25	20	47	0.6	1	—	4
40	—	**GE 40EC-2RS**	0.315	40	62	28	22	53	0.6	1	—	4
45	—	**GE 45EC-2RS**	0.413	45	68	32	25	60	0.6	1	—	4
50	—	**GE 50EC-2RS**	0.560	50	75	35	28	66	0.6	1	—	4
60	—	**GE 60EC-2RS**	1.10	60	90	44	36	80	1	1	—	3
70	—	**GE 70EC-2RS**	1.54	70	105	49	40	92	1	1	—	4

Mounting dimensions mm				Dynamic load capacity	Static load capacity
d_a		D_a		C_d	C_s
Min.	Max.(2)	Max.	Min.	N	N
17.5	18	23.5	21.5	19 400	48 500
19.5	20.5	27.5	24.5	24 500	61 300
22.5	24	32.5	28	34 100	85 300
29	29	37.5	34	55 700	139 000
34	34	42.5	41.5	71 800	180 000
39.5	39.5	49.5	48	92 200	230 000
44.5	45	56.5	54.5	114 000	286 000
49.5	50.5	62.5	60	147 000	368 000
54.5	56	69.5	66	181 000	453 000
65.5	66.5	84.5	79	282 000	706 000
75.5	77.5	99.5	91	361 000	902 000

Notes(1) Minimum allowable value of chamfer dimensions r_1 and r_2
(2) When Spherical Bushings are used with full tilting angle, the shaft shoulder dimension must be less than the maximum value of d_a.
Remark No oil hole is provided.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

PILLOBALLS

- ●PILLOBALL Spherical Bushings - Insert Type
- ●PILLOBALL Rod Ends - Insert Type
- ●PILLOBALL Rod Ends - Die-cast Type
- ●PILLOBALL Rod Ends - Maintenance-free Type

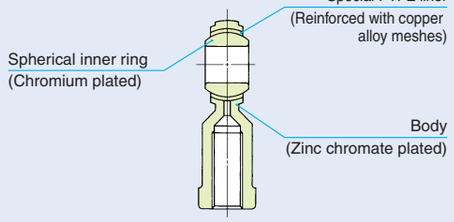


Structure and Features

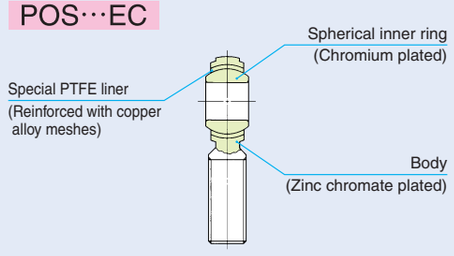
IKO PILLOBALLs are compact self-aligning spherical bushings that can support a large radial load and a bi-directional axial load at the same time.
These bushings are classified by sliding surface types, namely, insert type, die-cast type and maintenance-free type. In the insert type, a spherical inner ring makes contact with the special copper alloy bushing with superior run-in properties. In the die-cast type, a spherical inner ring makes direct contact with the bore surface of the body of special zinc die-cast alloy. In the maintenance-free type, a spherical inner ring makes contact with the special PTFE liner of maintenance-free type. Thus, a smooth rotational and oscillatory motion can be achieved with superior anti-wear and loading properties in each type.
PILLOBALL Rod Ends have either a female thread in the body or a male thread on the body, and they can be easily assembled onto machines.
PILLOBALLs are used in control and link mechanisms in machine tools, textile machines, packaging machines, etc. The maintenance-free type is especially suitable for loading in one direction and is the best choice for machines in which oil must be avoided such as food processing machines, or machines which cannot be re-lubricated.

Structures of maintenance-free type PILLOBALLs

PHS···EC

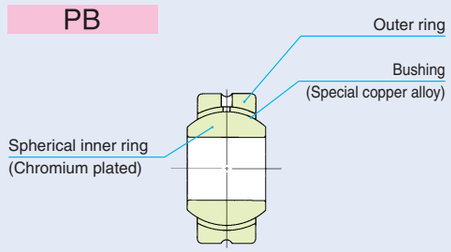


POS···EC

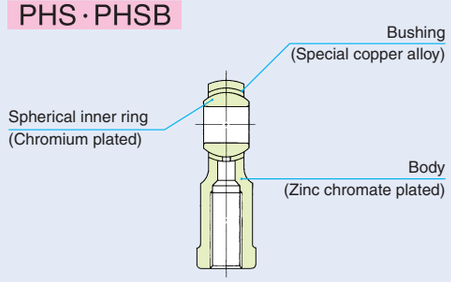


Structures of lubrication type PILLOBALLs

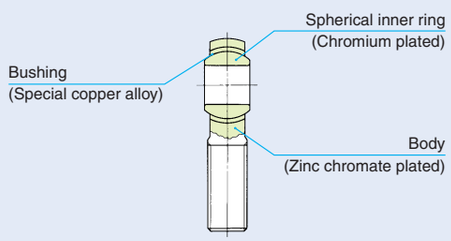
PB



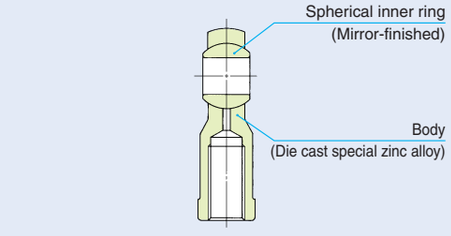
PHS·PHSB



POS·POSB



PHSA



Types

In PILLOBALLs, the types shown in Table 1 are available.

Table 1 Type

Type	Lubrication type			Maintenance-free type	
	Spherical Bushings	Rod end		Rod end	
		female thread	male thread	female thread	male thread
Insert type	PB	PHS·PHSB	POS·POSB	PHS···EC	POS···EC
Die-cast type	—	PHSA	—		

Lubrication Type PILLOBALL Spherical Bushings Insert Type PB

This type has superior anti-wear properties and high rigidity. It consists of a spherical inner ring, an outer ring, and a bushing of special copper alloy with superior run-in properties inserted in between. The spherical surface of the inner ring is chromium plated after heat treatment and grinding. This type is assembled with a shaft and a housing.
When especially large radial and/or axial loads are applied, Spherical Bushings with molybdenum disulfide (MoS_2) treated inner and outer rings are recommended. (See page J17.)

Lubrication Type PILLOBALL Rod Ends Insert Type PHS, POS, PHSB and POSB

This type has superior anti-wear and anti-corrosion properties as well as high rigidity. It consists of a spherical inner ring of which spherical surface is chromium-plated after heat treatment and grinding, a body with a zinc chromate treated outer surface, and an inserted bushing of special copper alloy having superior run-in properties. This type includes PHS and PHSB, which has a female thread in the body, and POS and POSB, which has a male thread on the body

Lubrication Type PILLOBALL Rod Ends Die-cast Type PHSA

The spherical inner ring of this type is mirror-finished after heat treatment and is built in a body of die-cast special zinc alloy. The sliding surfaces of the inner ring and body are in close contact with each other. Thus, this type is an economical rod end with superior anti-wear and loading properties.

Maintenance-free Type PILLOBALL Rod Ends PHS···EC, POS···EC

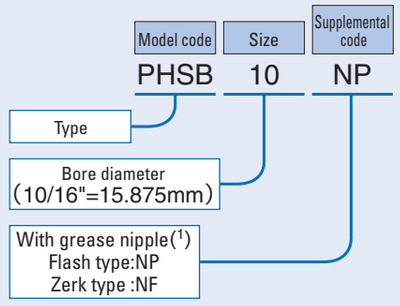
This type has superior anti-corrosion properties as the body is zinc chromate treated and the spherical inner ring is chromium plated on the sphere surface after heat treatment and grinding.
A special PTFE liner, reinforced with copper alloy meshes, which is superior in anti-wear properties with little creep deformation is used for lining on the sliding surface of the body, and this type is maintenance-free.
PHS···EC, which has a female thread in the body, and POS ···EC, which has a male thread on the body, are available.

Identification number

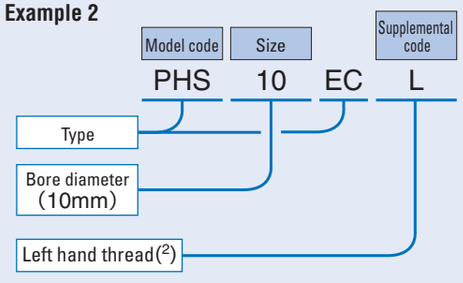
The identification number of PILLOBALLs consists of a model code, a size and any supplemental codes as shown in the examples.

Examples of identification number

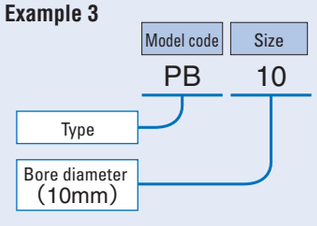
Example 1



Example 2



Example 3



Notes(1) Shapes of greace nipple are shown in Fig.1. In case of no indication of grease nipple type, zerk type is supplied.
(2) Right hand thread is indicated with no code.

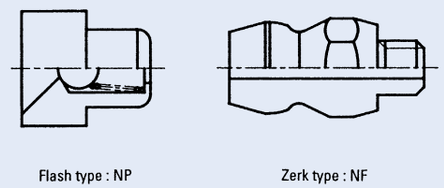


Fig. 1 Shapes of grease nipple

Accuracy

The accuracy of PILLOBALLs is shown in Tables 2 and 3. The maximum radial internal clearance of the insert type is 0.035 mm.

Table 2 Tolerance unit: mm

Type	Dimension	Dimension symbol	Tolerance
PB	Bore dia. of inner ring	d	H7
	Outside dia. of outer ring	D	h6
	Width of inner ring	B	0 / −0.1
	Width of outer ring	C	±0.1
PHS POS PHS···EC POS···EC	Bore dia. of inner ring	d	H7
	Width of inner ring	B	0 / −0.1
PHSB POSB	Bore dia. of inner ring	d	+0.038 / −0.013
	Width of inner ring	B_1	0 / −0.127
PHSA	Bore dia. of inner ring	d	+0.063 / −0.012
	Width of inner ring	B	See Table 3.

Table 3 Tolerance of width B of inner ring of PHSA type unit: mm

d Nominal bore dia. of inner ring		Δ_{Bs} Deviation of a single inner ring width	
Over	Incl.	High	Low
—	14	0	−0.2
14	20	0	−0.3
20	22	0	−0.4

Fit

Recommended fits for PILLOBALLs are shown in Table 4.

Table 4 Recommended fits

Condition	Tolerance class	
	Shaft	Housing bore(1)
Normal operation	h7	H7
Directionally indeterminate loading	n6, p6	N7

Note(1) This is applicable to PILLOBALL Spherical Bushings, Insert type.

Selection of PILLOBALL

Load capacities of PILLOBALLs are determined based on the allowable contact pressure on sliding surfaces and the strength of body for each type. Thus, a suitable type and size should be selected based on the dynamic load capacity C_d and static load capacity C_s shown in the dimension tables.

Load capacity

❶ Dynamic load capacity
The dynamic load capacity C_d is obtained on the basis of the contact pressure on the sliding surface. The dynamic load capacity is used for calculating the life.
The dynamic load capacity considering temperature increase is obtained from the following equation using the temperature factor, which is a correction factor for the effect of PILLOBALL temperature.

$$C_{dt} = f_t C_d \quad \cdots\cdots(1)$$

where, C_{dt} : Dynamic load capacity considering temperature increase, N
f_t : Temperature factor (Refer to Table 5.)
C_d : Dynamic load capacity, N (Refer to the dimension tables.)

Table 5 Temperature factor f_t

Type	Temperature ℃					
	−30 +80	+80 +90	+90 +100	+100 +120	+120 +150	+150 +180
PB PHS, POS PHSB, POSB	1	1	1	1	1	0.7
PHS···EC POS···EC	1	1	0.9	0.75	0.55	—

❷ Static load capacity
The static load capacity C_s is the maximum static load that can be applied on the PILLOBALL without breaking the inner or outer ring of the PILLOBALL Spherical Bushing (or the inner ring or body of the PILLOBALL Rod End), and without causing severe permanent deformation that will make the PILLOBALL unusable.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

Maximum Operating Load

The recommended value of bushing load is obtained by multiplying the dynamic load capacity C_d by a numerical factor, which differs depending on the bushing type and load condition. For PILLOBALL Rod Ends, the static load capacity C_s must also be considered in determining the applicable bushing load.
Table 6 shows the guidelines for maximum operating load of PILLOBALLs. When axial loads are added in addition to radial loads, bending stress occurs in the body. Pay attention to this bending stress.

Table 6 Maximum operating load

Type	Load direction	
	Constant	Alternate
PB	$\leqq 0.3C_d$ ($\leqq C_s$)	$\leqq 0.6C_d$
PHS,POS,PHSB,POSB	$\leqq 0.3C_d$ ($\leqq 0.3C_s$)	($\leqq 0.6C_d$) $\leqq 0.2C_s$
PHSA	$\leqq 0.16C_s$	
PHS···EC,POS···EC	($\leqq C_d$) $\leqq 0.3C_s$	($\leqq 0.5C_d$) $\leqq 0.2C_s$

Remark C_d is the dynamic load capacity and C_s is the static load capacity.
When the magnitude of applied load is within the value shown outside the parenthesis, it is also within the value in the parenthesis.

Equivalent radial load

PILLOBALLs can take radial and axial loads at the same time. When the magnitude and direction of loads are constant, the equivalent radial load can be obtained by the following formula.

$$P = F_r + YF_a \quad \cdots\cdots(2)$$

where, P : Equivalent radial load, N
F_r : Radial load, N
F_a : Axial load, N
Y : Axial load factor (Refer to Table 7.)

Table 7 Axial load factor Y

Type \ F_a/F_r	0.1	0.2	0.3	0.4	0.5	>0.5
PB PHS,POS PHSB,POSB	1	2	3	4	5	Unusable
PHS···EC POS···EC	1	2	3	Unusable		

Life

The life of PILLOBALLs is defined as the total number of oscillating motions during which the PILLOBALLs can be operated without failure or malfunction due to wear, increase in internal clearance, increase in sliding torque and operating temperature, etc.

As the actual life is affected by many factors such as the material of the sliding surface, the magnitude and direction of load, lubrication, sliding velocity, etc., the calculated life can be used as a measure of expected service life.

❶ Life of lubrication type PILLOBALLs PB · PHS · POS · PHSB · POSB

[1] Confirmation of pV value

Before attempting to calculate the life, make sure that the operating conditions are within the permissible range by referring to the pV diagram in Fig.2.
When the operating conditions are out of the permissible range, please consult IKO.
The contact pressure p and the sliding velocity V are obtained from the following formulae.

$$p = \frac{50P}{C_{dt}} \quad \cdots\cdots(3)$$

$$V = 5.82 \times 10^{-4} d_k \beta f \quad \cdots\cdots(4)$$

where, p : Contact pressure, N/mm²
P : Equivalent radial load, N
(Refer to Formula (2).)
C_{dt} : Dynamic load capacity considering temperature increase, N
(Refer to Formula (1).)
V : Sliding velocity, mm/s
d_k : Sphere diameter, mm
(Refer to the dimensional tables.)
2β : Oscillating angle degrees (Refer to Fig.2.)
when $\beta < 5°$, $\beta = 5$
when rotating, $\beta = 90$
f : Number of oscillations per minute, cpm

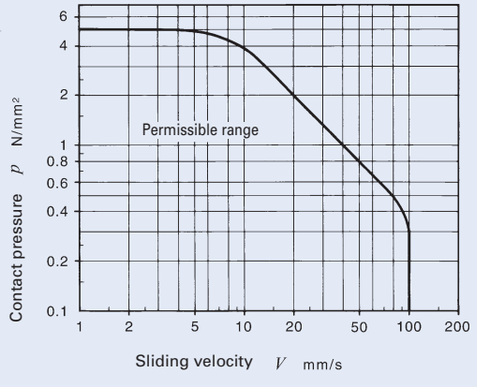


Fig. 2 pV diagram of lubrication type PILLOBALLs

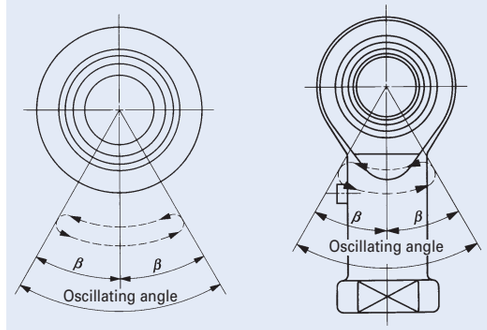


Fig. 3 Oscillating angle

[2] Life calculation

The life of lubrication type PILLOBALLs can be calculated by the following formulae.

$$G = \frac{3.18 b_1 b_2 b_3}{\sqrt{d_k \beta}} \left(\frac{C_{dt}}{P}\right)^2 \times 10^5 \quad \cdots\cdots(5)$$

$$L_h = \frac{G}{60f} \quad \cdots\cdots(6)$$

where, G : Life (Total number of oscillations)
b_1 : Load directional factor (Refer to Table 8.)
b_2 : Lubrication factor (Refer to Table 8.)
b_3 : Sliding velocity factor (Refer to Fig. 3.)
C_{dt} : Dynamic load capacity considering temperature increase, N
(Refer to Formula (1).)
P : Equivalent radial load, N
(Refer to Formula (2).)
L_h : Life in hours, h
f : Number of oscillations per minute, cpm

Table 8 Load directional factor b_1 and lubrication factor b_2 for lubrication type PILLOBALLs

Load directional factor b_1		Lubrication factor b_2	
Load direction		Periodical lubrication	
Constant	Alternate	None	Regular
1	5	1	15

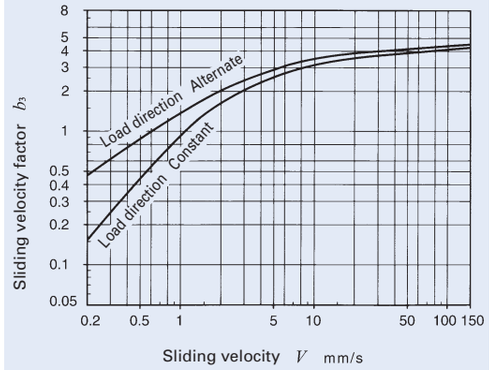


Fig. 4 Sliding velocity factor for lubrication type PILLOBALLs

❷ Life of maintenance-free type PILLOBALLs PHS···EC · POS···EC

[1] Confirmation of pV value

Before attempting to calculate the life, make sure that the operating conditions are within the permissible range by referring to the pV diagram in Fig.4.
When the operating conditions are out of the permissible range, please consult IKO.
The contact pressure p and sliding velocity V are obtained from Formulae (3) and (4) on page K6.

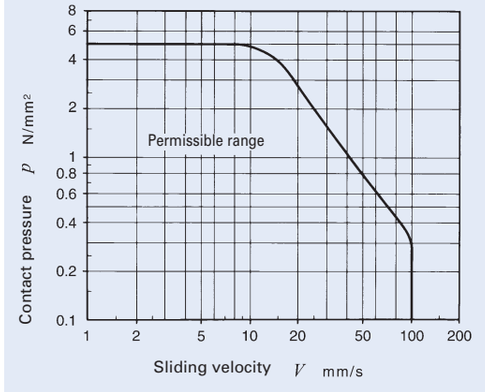


Fig. 5 pV diagram for maintenance-free type PILLOBALL Rod Ends

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

[2] Life calculation

The life of maintenance-free type PILLOBALL Rod Ends is obtained from the total sliding distance S which is given in Fig.5 for the contact pressure p obtained from Formula (3).

The total number of oscillations and life in hours can be obtained from the following formulae.

$$G = 16.67 \times b_1 \times \frac{Sf}{V} \quad \cdots\cdots(7)$$

$$L_h = \frac{G}{60f} \quad \cdots\cdots(8)$$

where, G : Life (Total number of oscillations)
b_1 : Load directional factor (Refer to Table 9.)
S : Total sliding distance m
f : Number of oscillations per minute cpm
V : Sliding velocity mm/s
L_h : Life in hours h

Table 9 Load directional factor for maintenance-free type PILLOBALLs b_1

Load direction	Constant	Alternate
Load directional factor b_1	1	0.2(1)

Note(1) This value is applicable when the load changes comparatively slowly. When the load changes rapidly, please consult IKO, as the factor degreases sharply.

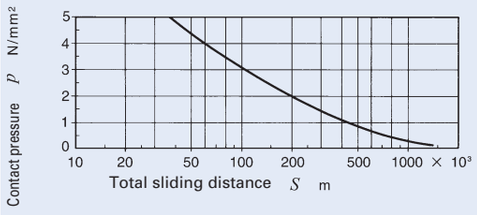


Fig. 6 Contact pressure and total sliding distance for maintenance-free type PILLOBALL Rod Ends

Lubrication

Maintenance-free type PILLOBALL Rod Ends have a sliding surface lined with a self-lubricating lining. Therefore, they can be used without lubrication.

Lubrication type PILLOBALLs are not provided with prepacked grease. Perform proper lubrication for use. Operating without lubrication will increase the wear of the sliding contact surfaces and cause seizure.

Oil Hole and Grease Nipple

Table 10 shows the specifications of oil hole and grease nipple on the outer ring or body. When a grease gun that fits the grease nipple is required, please contact IKO.

For PILLOBALLs without an oil hole and grease nipple, apply grease directly on the spherical surface.

Table 10 Specifications of oil hole and grease nipple

Type	Bore diameter of inner ring d mm	Specification
PB		1 oil hole + oil groove
PHS	$d \leqq 4$	None
	$4 < d$	With grease nipple
POS	$d \leqq 4$	None
	$4 < d \leqq 6$	1 oil hole
	$6 < d$	With grease nipple
PHSA		With grease nipple
PHS ··· EC, POS ··· EC		None

Operating Temperature Range

The maximum allowable temperature for Lubrication type PILLOBALLs is +180˚C for the insert type and +80˚C for the die-cast type.

The maximum allowable temperature for Maintenance-free type PILLOBALL Rod Ends is +150˚C.

Precautions for Use

❶ Tightening depth

The recommended tightening depth of the screw into the PILLOBALL Rod End body is shown below.

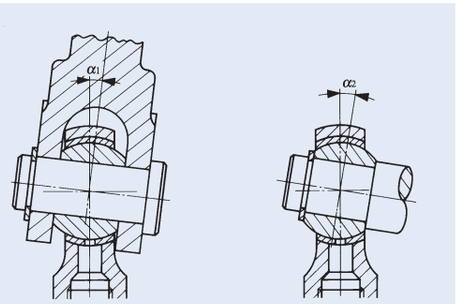
Insert type and maintenance-free type: 1.25 times the nominal thread dia. or more.

Die-cast type: 2 times the nominal thread dia. or more.

❷ Allowable tilting angle

The allowable tilting angle differs depending on the mounting structure as shown in Table 11.

Table 11 Allowable tilting angle



unit: degree

d Bore diameter mm	PB(1), PHS, POS PHS ··· EC, POS ··· EC		PHSA	
	α_1	α_2	α_1	α_2
3	7	13	—	—
4	7	13	—	—
5	8	13	7	13
6	8	13	7	13
8	8	14	8	14
10	8	14	8	14
12	8	13	8	13
14	10	16	9	16
16	9	15	9	15
18	9	15	9	15
20	9	15	9	15
22	10	15	9	15
25	9	15	—	—
28	9	15	—	—
30	10	17	—	—

Note(1) In the case of the PB series, α_2 is applicable in general.

Table 12 Allowable tilting angle for inch series

unit: degree

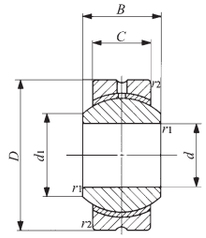
With female thread	With male thread	α_1	α_2
PHSB 2	POSB 2	8	16
PHSB 2.5	POSB 2.5	7	12
PHSB 3	POSB 3	6	10
PHSB 4	POSB 4	7	13
PHSB 5	POSB 5	6	10
PHSB 6	POSB 6	6	11
PHSB 7	POSB 7	7	11
PHSB 8	POSB 8	6	9
PHSB 10	POSB 10	7	11
PHSB 12	POSB 12	6	10
PHSB 16	POSB 16	7	14

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

PILLOBALL

Lubrication Type PILLOBALL Spherical Bushings Insert Type





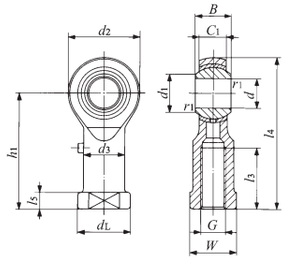
PB

Identification number	Mass (Ref.) g	Boundary dimensions mm d	D	C	B	d_1	$r_{s\,min}$ (1)	Ball dia. mm (inch)	Dynamic load capacity C_d N	Static load capacity C_s N
PB 5	8.5	5	16	6	8	7.7	0.2	11.112 (7/16)	3 270	7 850
PB 6	13	6	18	6.75	9	9	0.2	12.700 (1/2)	4 200	10 100
PB 8	24	8	22	9	12	10.4	0.2	15.875 (5/8)	7 010	16 800
PB 10	39	10	26	10.5	14	12.9	0.2	19.050 (3/4)	9 810	23 500
PB 12	58	12	30	12	16	15.4	0.2	22.225 (7/8)	13 100	31 400
PB 14	84	14	34	13.5	19	16.9	0.3	25.400 (1)	16 800	40 400
PB 16	111	16	38	15	21	19.4	0.3	28.575 (1 1/8)	21 000	50 400
PB 18	160	18	42	16.5	23	21.9	0.3	31.750 (1 1/4)	25 700	61 600
PB 20	210	20	46	18	25	24.4	0.3	34.925 (1 3/8)	30 800	74 000
PB 22	265	22	50	20	28	25.8	0.3	38.100 (1 1/2)	37 400	89 700
PB 25	390	25	56	22	31	29.6	0.6	42.862 (1 11/16)	46 200	111 000
PB 28	410	28	62	25	35	32.3	0.6	47.625 (1 7/8)	58 400	140 000
PB 30	610	30	66	25	37	34.8	0.6	50.800 (2)	62 300	149 000

Note(1) Minimum allowable value of chamfer dimensions r_1 and r_2
Remarks1. The outer ring has an oil groove and an oil hole.
2. Not provided with prepacked grease. Perform proper lubrication for use.

Lubrication Type PILLOBALL Rod Ends Insert Type/With Female Thread





PHS

Identification number	Mass (Ref.) g	Boundary dimensions mm d	Thread G	d_2	C_1	B	d_1	l_4	h_1	l_3	l_5	W	d_3	d_L	$r_{1s\,min}$ (1)	Ball dia. mm (inch)	Dynamic load capacity C_d N	Static load capacity C_s N
PHS 3	5.7	3	M 3×0.5	12	4.5	6	5.2	27	21	10	3	5.5	5	6.5	0.2	7.938 (5/16)	1 750	3 670
PHS 4	11.9	4	M 4×0.7	14	5.3	7	6.5	31	24	12	4	8	8	9.5	0.2	9.525 (3/8)	2 480	4 680
PHS 5	16.5	5	M 5×0.8	16	6	8	7.7	35	27	14	4	9	9	11	0.2	11.112 (7/16)	3 270	5 730
PHS 6	25	6	M 6×1	18	6.75	9	9	39	30	14	5	11	10	13	0.2	12.700 (1/2)	4 200	6 910
PHS 8	43	8	M 8×1.25	22	9	12	10.4	47	36	17	5	14	12.5	16	0.2	15.875 (5/8)	7 010	10 200
PHS 10	72	10	M10×1.5	26	10.5	14	12.9	56	43	21	6.5	17	15	19	0.2	19.050 (3/4)	9 810	13 300
PHS 12	107	12	M12×1.75	30	12	16	15.4	65	50	24	6.5	19	17.5	22	0.2	22.225 (7/8)	13 100	16 900
PHS 14	160	14	M14×2	34	13.5	19	16.9	74	57	27	8	22	20	25	0.2	25.400 (1)	16 800	20 900
PHS 16	210	16	M16×2	38	15	21	19.4	83	64	33	8	22	22	27	0.2	28.575 (1 1/8)	21 000	25 400
PHS 18	295	18	M18×1.5	42	16.5	23	21.9	92	71	36	10	27	25	31	0.2	31.750 (1 1/4)	25 700	30 200
PHS 20	380	20	M20×1.5	46	18	25	24.4	100	77	40	10	30	27.5	34	0.2	34.925 (1 3/8)	30 800	35 500
PHS 22	490	22	M22×1.5	50	20	28	25.8	109	84	43	12	32	30	37	0.2	38.100 (1 1/2)	37 400	41 700
PHS 25	750	25	M24×2	60	22	31	29.6	124	94	48	12	36	33.5	42	0.6	42.862 (1 11/16)	46 200	72 700
PHS 28	950	28	M27×2	66	25	35	32.3	136	103	53	12	41	37	46	0.6	47.625 (1 7/8)	58 400	87 000
PHS 30	1 130	30	M30×2	70	25	37	34.8	145	110	56	15	41	40	50	0.6	50.800 (2)	62 300	92 200

Note(1) Minimum allowable value of chamfer dimension r_1
Remarks1. Neither oil hole nor grease nipple is provided for PHS with an inner ring bore diameter d of 4 mm or less. For others, a grease nipple is provided on the body.
2. Not provided with prepacked grease. Perform proper lubrication for use.
3. When a metric fine thread specification is required, please contact IKO.

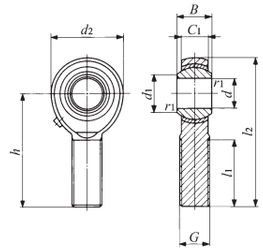
K
PB
PHS
PHSB
POS
POSB
PHSA

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

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Lubrication Type PILLOBALL Rod Ends Insert Type/With Male Thread





POS

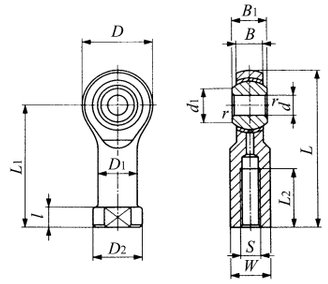
Identification number	Mass (Ref.)	Boundary dimensions mm										Dynamic load capacity	Static load capacity
		d	Thread G	d_2	C_1	B	d_1	l_2	h	l_1	$r_{1s\,min}$ (1) Ball dia. mm (inch)	C_d	C_s
	g											N	N
POS 3	5.0	3	M 3×0.5	12	4.5	6	5.2	33	27	15	0.2	7.938 (5/16)	1 750	1 220
POS 4	8.1	4	M 4×0.7	14	5.3	7	6.5	37	30	17	0.2	9.525 (3/8)	2 480	2 060
POS 5	12.5	5	M 5×0.8	16	6	8	7.7	41	33	20	0.2	11.112 (7/16)	3 270	3 340
POS 6	19	6	M 6×1	18	6.75	9	9	45	36	22	0.2	12.700 (1/2)	4 200	4 730
POS 8	32	8	M 8×1.25	22	9	12	10.4	53	42	25	0.2	15.875 (5/8)	7 010	8 640
POS 10	54	10	M10×1.5	26	10.5	14	12.9	61	48	29	0.2	19.050 (3/4)	9 810	13 300
POS 12	85	12	M12×1.75	30	12	16	15.4	69	54	33	0.2	22.225 (7/8)	13 100	16 900
POS 14	126	14	M14×2	34	13.5	19	16.9	77	60	36	0.2	25.400 (1)	16 800	20 900
POS 16	185	16	M16×2	38	15	21	19.4	85	66	40	0.2	28.575 (1 1/8)	21 000	25 400
POS 18	260	18	M18×1.5	42	16.5	23	21.9	93	72	44	0.2	31.750 (1 1/4)	25 700	30 200
POS 20	340	20	M20×1.5	46	18	25	24.4	101	78	47	0.2	34.925 (1 3/8)	30 800	35 500
POS 22	435	22	M22×1.5	50	20	28	25.8	109	84	51	0.2	38.100 (1 1/2)	37 400	41 700
POS 25	650	25	M24×2	60	22	31	29.6	124	94	57	0.6	42.862 (1 11/16)	46 200	72 700
POS 28	875	28	M27×2	66	25	35	32.3	136	103	62	0.6	47.625 (1 7/8)	58 400	87 000
POS 30	1 070	30	M30×2	70	25	37	34.8	145	110	66	0.6	50.800 (2)	62 300	92 200

Note(1) Minimum allowable value of chamfer dimension r_1

Remarks1. Neither oil hole nor grease nipple is provided for POS with an inner ring bore diameter d of 4 mm or less.
For those with an inner ring bore diameter d of 5 to 6 mm, an oil hole is provided on the body. For others, a grease nipple is provided on the body.
2. Not provided with prepacked grease. Perform proper lubrication for use.
3. When a metric fine thread specification is required, please contact IKO.

Inch series PILLOBALL Rod Ends Insert Type/With Female Thread





PHSB

Identification number	Mass (Ref.)	Boundary dimensions mm(inch)													Dynamic load capacity	Static load capacity
		d	Thread S class 3B	D	B	B_1	d_1	L	l	L_1	L_2	W	D_1	D_2	$r_{s\,min}$ (1) Ball dia. mm (inch)	C_d	C_s
	g															N	N
PHSB 2	6.8	3.175 (.1250)	-32UNC (.1380)	11.91 (.469)	4.75 (.187)	6.35 (.250)	4.75 (.187)	26.57 (1.046)	4.75 (.187)	20.62 (.812)	9.53 (.375)	6.35 (.250)	6.35 (.250)	7.92 (.312)	0.3 (.012) 7.938 (5/16)	1 850	5 840
PHSB 2.5	11	3.967 (.1562)	-32UNC (.1640)	14.27 (.562)	5.56 (.219)	7.14 (.281)	6.32 (.249)	29.36 (1.156)	4.75 (.187)	22.23 (.875)	9.53 (.375)	7.14 (.281)	7.14 (.281)	8.74 (.344)	0.3 (.012) 9.525 (3/8)	2 600	8 210
PHSB 3	14	4.826 (.1900)	-32UNF (.1900)	15.88 (.625)	6.35 (.250)	7.92 (.312)	7.77 (.306)	34.93 (1.375)	4.75 (.187)	26.97 (1.062)	14.27 (.562)	7.92 (.312)	7.92 (.312)	10.31 (.406)	0.3 (.012) 11.112 (7/16)	3 460	9 090
PHSB 4	23	6.350 (.2500)	-28UNF (.2500)	19.05 (.750)	7.14 (.281)	9.53 (.375)	9.02 (.355)	42.85 (1.687)	4.75 (.187)	33.32 (1.312)	19.05 (.750)	9.53 (.375)	9.53 (.375)	11.89 (.468)	0.5 (.020) 13.097 (33/64)	4 590	13 200
PHSB 5	36	7.938 (.3125)	-24UNF (.3125)	22.23 (.875)	8.74 (.344)	11.10 (.437)	11.35 (.447)	46.02 (1.812)	4.75 (.187)	34.93 (1.375)	19.05 (.750)	11.10 (.437)	11.10 (.437)	12.70 (.500)	0.5 (.020) 15.875 (5/8)	6 800	16 500
PHSB 6	59	9.525 (.3750)	-24UNF (.3750)	25.40 (1.000)	10.31 (.406)	12.70 (.500)	13.13 (.517)	53.98 (2.125)	6.35 (.250)	41.28 (1.625)	23.80 (.937)	14.27 (.562)	14.27 (.562)	17.45 (.687)	0.5 (.020) 18.256 (23/32)	9 230	21 600
PHSB 7	82	11.112 (.4375)	-20UNF (.4375)	28.58 (1.125)	11.10 (.437)	14.27 (.562)	14.88 (.586)	60.33 (2.375)	6.35 (.250)	46.02 (1.812)	26.97 (1.062)	15.88 (.625)	15.88 (.625)	19.05 (.750)	0.5 (.020) 20.638 (13/16)	11 200	26 100
PHSB 8	132	12.700 (.5000)	-20UNF (.5000)	33.32 (1.312)	12.70 (.500)	15.88 (.625)	17.73 (.698)	70.64 (2.781)	6.35 (.250)	53.98 (2.125)	30.15 (1.187)	19.05 (.750)	19.05 (.750)	22.23 (.875)	0.5 (.020) 23.812 (15/16)	14 800	36 200
PHSB 10	191	15.875 (.6250)	-18UNF (.6250)	38.10 (1.500)	14.27 (.562)	19.05 (.750)	21.31 (.839)	82.55 (3.250)	7.92 (.312)	63.50 (2.500)	38.10 (1.500)	22.23 (.875)	22.23 (.875)	25.40 (1.000)	0.5 (.020) 28.575 (1 1/8)	20 000	39 300
PHSB 12	286	19.050 (.7500)	-16UNF (.7500)	44.45 (1.750)	17.45 (.687)	22.23 (.875)	24.84 (.978)	95.25 (3.750)	7.92 (.312)	73.03 (2.875)	44.45 (1.750)	25.40 (1.000)	25.40 (1.000)	28.58 (1.125)	0.5 (.020) 33.338 (1 5/16)	28 500	55 000
PHSB 16	998	25.400 (1.0000)	-12UNF (1.2500)	69.85 (2.750)	25.40 (1.000)	34.93 (1.375)	32.23 (1.269)	139.70 (5.500)	11.07 (.436)	104.78 (4.125)	53.98 (2.125)	38.10 (1.500)	38.10 (1.500)	44.45 (1.750)	0.5 (.020) 47.625 (1 7/8)	59 300	86 800

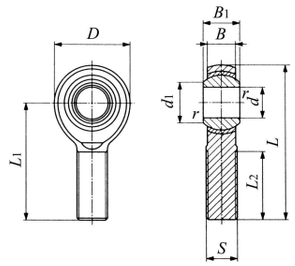
Note(1) r_s min stands for minimum allowable value of chamfer r.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

PILLOBALL

Inch series PILLOBALL Rod Ends Insert Type/With Male Thread





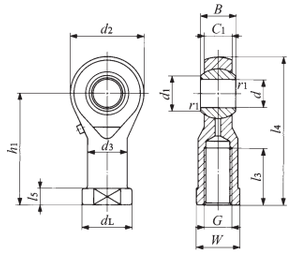
POSB

Identification number	Mass (Ref.) g	Boundary dimensions mm(inch) d	Thread S class 3A	D	B	B_1	d_1	L	L_1	L_2	$r_{s\,min}$ (1)	Ball dia. mm (inch)	Dynamic load capacity C_d N	Static load capacity C_s N
POSB 2	5.4	3.175 (.1250)	-32UNC (.1380)	11.91 (.469)	4.75 (.187)	6.35 (.250)	4.75 (.187)	29.77 (1.172)	23.80 (.937)	12.70 (.500)	0.3 (.012)	7.938 (5/16)	1 850	2 160
POSB 2.5	9.1	3.967 (.1562)	-32UNC (.1640)	14.27 (.562)	5.56 (.219)	7.14 (.281)	6.32 (.249)	35.71 (1.406)	28.58 (1.125)	15.88 (.625)	0.3 (.012)	9.525 (3/8)	2 600	3 370
POSB 3	14	4.826 (.1900)	-32UNF (.1900)	15.88 (.625)	6.35 (.250)	7.92 (.312)	7.77 (.306)	39.70 (1.563)	31.75 (1.250)	19.05 (.750)	0.3 (.012)	11.112 (7/16)	3 460	4 850
POSB 4	23	6.350 (.2500)	-28UNF (.2500)	19.05 (.750)	7.14 (.281)	9.53 (.375)	9.02 (.355)	49.20 (1.937)	39.67 (1.562)	25.40 (1.000)	0.5 (.020)	13.097 (33/64)	4 590	8 870
POSB 5	36	7.938 (.3125)	-24UNF (.3125)	22.23 (.875)	8.74 (.344)	11.10 (.437)	11.35 (.447)	58.72 (2.312)	47.63 (1.875)	31.75 (1.250)	0.5 (.020)	15.875 (5/8)	6 800	14 200
POSB 6	54	9.525 (.3750)	-24UNF (.3750)	25.40 (1.000)	10.31 (.406)	12.70 (.500)	13.13 (.517)	61.93 (2.438)	49.23 (1.938)	31.75 (1.250)	0.5 (.020)	18.256 (23/32)	9 230	21 600
POSB 7	77	11.112 (.4375)	-20UNF (.4375)	28.58 (1.125)	11.10 (.437)	14.27 (.562)	14.88 (.586)	68.28 (2.688)	53.98 (2.125)	34.93 (1.375)	0.5 (.020)	20.638 (13/16)	11 200	26 100
POSB 8	122	12.700 (.5000)	-20UNF (.5000)	33.32 (1.312)	12.70 (.500)	15.88 (.625)	17.73 (.698)	78.59 (3.094)	61.93 (2.438)	38.10 (1.500)	0.5 (.020)	23.812 (15/16)	14 800	36 200
POSB 10	186	15.875 (.6250)	-18UNF (.6250)	38.10 (1.500)	14.27 (.562)	19.05 (.750)	21.31 (.839)	85.73 (3.375)	66.68 (2.625)	41.28 (1.625)	0.5 (.020)	28.575 (1 1/8)	20 000	39 300
POSB 12	295	19.050 (.7500)	-16UNF (.7500)	44.45 (1.750)	17.45 (.687)	22.23 (.875)	24.84 (.978)	95.25 (3.750)	73.03 (2.875)	44.45 (1.750)	0.5 (.020)	33.338 (1 5/16)	28 500	55 000
POSB 16	1 129	25.400 (1.0000)	-12UNF (1.2500)	69.85 (2.750)	25.40 (1.000)	34.93 (1.375)	32.23 (1.269)	139.70 (5.500)	104.78 (4.125)	53.98 (2.125)	0.5 (.020)	47.625 (1 7/8)	59 300	112 000

Note(1) r_s min stands for minimum allowable value of chamfer r.

Lubrication Type PILLOBALL Rod Ends Die-cast Type/With Female Thread





PHSA

Identification number	Mass (Ref.) g	Boundary dimensions mm d	Thread G	d_2	C_1	B	d_1	l_4	h_1	l_3	l_5	W	d_3	d_L	$r_{1s\,min}$ (1)	Ball dia. mm (inch)	Static load capacity C_s N
PHSA 5	17	5	M 5×0.8	17	6	8	7.7	35.5	27	16	4	9	9	11	0.2	11.112 (7/16)	5 470
PHSA 6	25	6	M 6×1	19.5	6.75	9	9	39.7	30	16	5	11	10	13	0.2	12.700 (1/2)	6 760
PHSA 8	45	8	M 8×1.25	24	9	12	10.4	48	36	19	5	14	12.5	16	0.2	15.875 (5/8)	10 200
PHSA 10	70	10	M10×1.5	28	10.5	14	12.9	57	43	23	6.5	17	15	19	0.2	19.050 (3/4)	13 100
PHSA 12	105	12	M12×1.75	32	12	16	15.4	66	50	27	6.5	19	17.5	22	0.2	22.225 (7/8)	16 400
PHSA 14	155	14	M14×2	36	13.5	19	16.9	75	57	30	8	22	20	25	0.3	25.400 (1)	20 000
PHSA 16	190	16	M16×2	40	15	21	19.4	84	64	36	8	22	22	27	0.3	28.575 (1 1/8)	23 900
PHSA 18	290	18	M18×1.5	45	16.5	23	21.9	93.5	71	40	10	27	25	31	0.3	31.750 (1 1/4)	28 800
PHSA 20	400	20	M20×1.5	49	18	25	24.4	101.5	77	43	10	30	27.5	34	0.3	34.925 (1 3/8)	33 400
PHSA 22	500	22	M22×1.5	54	20	28	25.8	111	84	47	12	32	30	37	0.3	38.100 (1 1/2)	40 400

Note(1) Minimum allowable value of chamfer dimension r_1

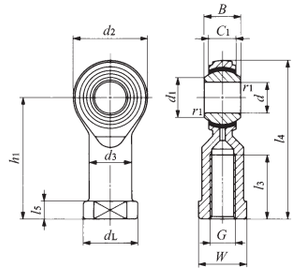
Remarks1. A grease nipple is provided on the body.
2. Not provided with prepacked grease. Perform proper lubrication for use.
3. When a metric fine thread specification is required, please contact IKO.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

PILLOBALL

Maintenance-free Type PILLOBALL Rod Ends With Female Thread





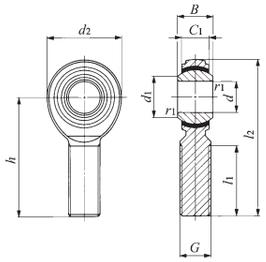
PHS···EC

Identification number	Mass (Ref.)	Boundary dimensions mm															Dynamic load capacity	Static load capacity
		d	Thread G	d_2	C_1	B	d_1	l_4	h_1	l_3	l_5	W	d_3	d_L	$r_{1s\,min}$ (1)	Ball dia. mm (inch)	C_d	C_s
	g																N	N
PHS 3EC	5.7	3	M 3×0.5	12	4.5	6	5.2	27	21	10	3	5.5	5	6.5	0.2	7.938 (5/16)	3 500	2 480
PHS 4EC	11.9	4	M 4×0.7	14	5.3	7	6.5	31	24	12	4	8	8	9.5	0.2	9.525 (3/8)	4 950	3 260
PHS 5EC	16.5	5	M 5×0.8	16	6	8	7.7	35	27	12.5	4	9	9	11	0.2	11.112 (7/16)	6 540	4 010
PHS 6EC	25	6	M 6×1	18	6.75	9	9	39	30	13.5	5	11	10	13	0.2	12.700 (1/2)	8 410	4 940
PHS 8EC	43	8	M 8×1.25	22	9	12	10.4	47	36	16	5	14	12.5	16	0.2	15.875 (5/8)	14 000	7 760
PHS 10EC	72	10	M10×1.5	26	10.5	14	12.9	56	43	19.5	6.5	17	15	19	0.2	19.050 (3/4)	19 600	10 500
PHS 12EC	107	12	M12×1.75	30	12	16	15.4	65	50	24	6.5	19	17.5	22	0.2	22.225 (7/8)	26 200	13 700
PHS 14EC	160	14	M14×2	34	13.5	19	16.9	74	57	27	8	22	20	25	0.2	25.400 (1)	33 600	17 200
PHS 16EC	210	16	M16×2	38	15	21	19.4	83	64	33	8	22	22	27	0.2	28.575 (1 1/8)	42 000	21 100
PHS 18EC	295	18	M18×1.5	42	16.5	23	21.9	92	71	36	10	27	25	31	0.2	31.750 (1 1/4)	51 400	25 100
PHS 20EC	380	20	M20×1.5	46	18	25	24.4	100	77	40	10	30	27.5	34	0.2	34.925 (1 3/8)	61 600	30 000
PHS 22EC	490	22	M22×1.5	50	20	28	25.8	109	84	41	12	32	30	37	0.2	38.100 (1 1/2)	74 700	36 400

Note(1) Minimum allowable value of chamfer dimension r_1
Remarks1. Neither oil hole nor grease nipple is provided.
2. When a metric fine thread specification is required, please contact IKO.

Maintenance-free Type PILLOBALL Rod Ends With Male Thread





POS···EC

Identification number	Mass (Ref.)	Boundary dimensions mm										Dynamic load capacity	Static load capacity
		d	Thread G	d_2	C_1	B	d_1	l_2	h	l_1	$r_{1s\,min}$ (1)	Ball dia. mm (inch)	C_d	C_s
	g												N	N
POS 3EC	5.0	3	M 3×0.5	12	4.5	6	5.2	33	27	15	0.2	7.938 (5/16)	3 500	1 220
POS 4EC	8.1	4	M 4×0.7	14	5.3	7	6.5	37	30	17	0.2	9.525 (3/8)	4 950	2 060
POS 5EC	12.5	5	M 5×0.8	16	6	8	7.7	41	33	20	0.2	11.112 (7/16)	6 540	3 340
POS 6EC	19	6	M 6×1	18	6.75	9	9	45	36	22	0.2	12.700 (1/2)	8 410	4 730
POS 8EC	32	8	M 8×1.25	22	9	12	10.4	53	42	25	0.2	15.875 (5/8)	14 000	7 760
POS 10EC	54	10	M10×1.5	26	10.5	14	12.9	61	48	29	0.2	19.050 (3/4)	19 600	10 500
POS 12EC	85	12	M12×1.75	30	12	16	15.4	69	54	33	0.2	22.225 (7/8)	26 200	13 700
POS 14EC	126	14	M14×2	34	13.5	19	16.9	77	60	36	0.2	25.400 (1)	33 600	17 200
POS 16EC	185	16	M16×2	38	15	21	19.4	85	66	40	0.2	28.575 (1 1/8)	42 000	21 100
POS 18EC	260	18	M18×1.5	42	16.5	23	21.9	93	72	44	0.2	31.750 (1 1/4)	51 400	25 100
POS 20EC	340	20	M20×1.5	46	18	25	24.4	101	78	47	0.2	34.925 (1 3/8)	61 600	30 000
POS 22EC	435	22	M22×1.5	50	20	28	25.8	109	84	51	0.2	38.100 (1 1/2)	74 700	36 400

Note(1) Minimum allowable value of chamfer dimension r_1
Remarks1. Neither oil hole nor grease nipple is provided.
2. When a metric fine thread specification is required, please contact IKO.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

L-BALLS

●L-Balls
●L-Ball Dust Cover



Structure and Features

IKO L-Balls are self-aligning rod-ends consisting of a special die-cast zinc alloy body and a studded ball which has its axis at right angles to the body.
They can perform tilting movement, oscillating movement and rotation with low torque, and transmit power smoothly due to uniform clearance between the sliding surfaces.
Their superior wear resistance assures stable accuracy for long periods of time, and maintenance is simple. They are very economical bearings.
For these reasons, they are widely used in link mechanisms in automobiles, construction machinery, farm and packaging machines, etc.

Types

IKO L-Balls are available in various types as shown in Table 1.

Table 1 Type of L-Balls

Type	L-Ball		L-Ball dust cover
Model code	LHSA	LHS	PRC

L-Ball LHSA

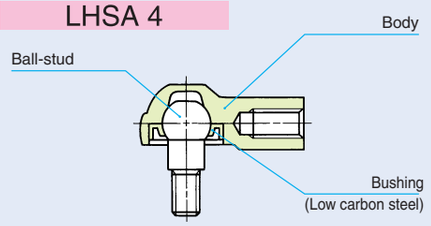
These are compact rod-ends in which the spherical part of the ball-stud are held by the special die-cast zinc alloy body. There is a dust cover on the stud side and good quality lithium soap base grease is prepacked. They can be run for long periods of time without re-lubrication and have excellent lubrication and anti-dust properties.
As shown in the structural drawing, these rod-ends are classified into 3 types by size. In addition, the ball-studs of LHSA 10 and lower are formed in one solid body, but those of LHSA 12 and higher, which are used under large loads, have the stud friction-welded to a high precision steel ball to give greater resistance to wear.

L-Ball LHS

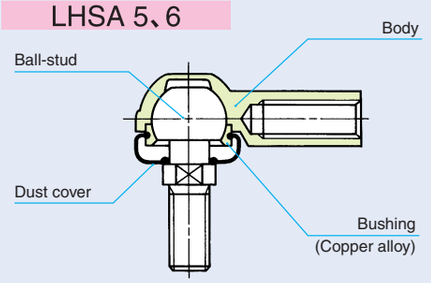
These rod-ends have a friction-welded ball-stud, and a special die-cast zinc alloy body which houses the spherical surface of the high precision steel ball. There is an almost complete contact across the sliding surfaces, and the uniform clearance guarantees a stable bearing life.
An L-Ball dust cover can be attached to these rod-ends. If the rod-ends are lubricated with lithium soap base grease, they have excellent lubrication and anti-dust properties and can run for long periods of time without re-lubrication.
When the L-Ball LHS is delivered with a dust cover on request, lithium soap base grease is prepacked.

Structures of L-Ball LHSA

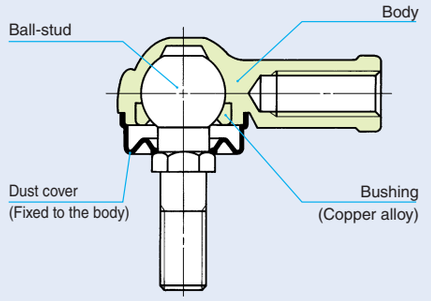
LHSA 4



LHSA 5、6

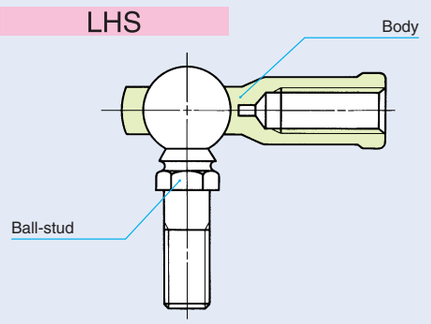


LHSA 8 and upwards



Structure of L-Ball LHS

LHS



K
LHSA
LHS

L-Ball Dust Cover PRC

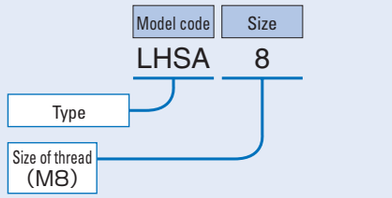
This is for the L-Ball LHS series. It is made of special synthetic rubber which has excellent resistance to oil and ozone. The cover offers very effective dust protection and prevents grease leakage.

Identification Number

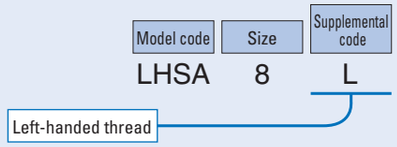
The identification number of L-Balls consists of a model code, a size and any supplemental codes as shown in the examples.

Examples of identification number

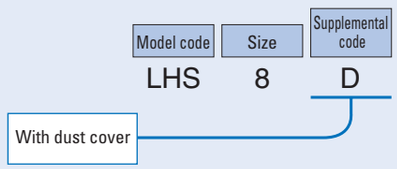
Example 1 (Female thread of the body : In case of right-hand threaded)



Example 2 (Female thread of the body : In case of left threaded)



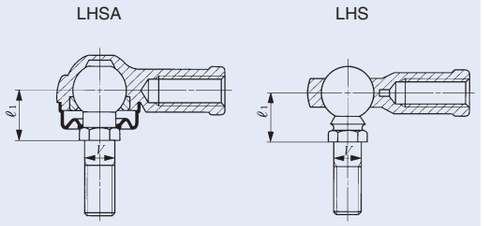
Example 3 (when a dust cover PRC is attached to LHS)



Accuracy

The accuracy of L-Balls is shown in Table 2.

Table 2 Tolerance



unit: mm

Type	Dimension symbol	Tolerance
LHSA	ℓ_1	±0.5
	V	0 −0.2(1)
LHS	ℓ_1	±0.4
	V	h9

Note(1) This dimensional tolerance applies to LHSA 5 and higher.

Selection of L-Balls

The static load capacity and maximum operating load of L-Balls are determined in consideration of the strength of the ball stud and the body. Accordingly, L-Balls are selected on the basis of the static load capacity C_s shown in the dimension table and the maximum operating load shown in Table 3.

Static load capacity

The static load capacity C_s shown in the dimension table represents the allowable axial force F which is determined by the mechanical strength of the ball-stud at the section 'A' under the bending moment due to the force F as illustrated in Fig. 1. If F increases beyond the static load capacity, deformation will begin at A, leading to breakage.

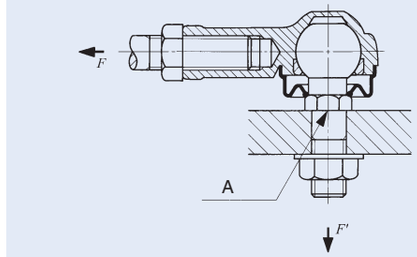


Fig. 1

Maximum operating load

The strength of the body must also be taken into consideration when L-Balls are operated in a high-temperature or low-temperature atmosphere or receive repetitive loads of long duration or shock loads. A guideline for maximum operating load of L-Balls is shown in Table 3. When the fixing bolt in the main body is fixed and a load is applied in the direction of F', the bending stress in the fixing bolt must be taken into consideration.

Table 3 Maximum operating load unit: N

Identification number	Maximum operating load	Identification number	Maximum operating load
LHSA 4	840	LHS 5	880
LHSA 5	1 180	LHS 6	1 080
LHSA 6	1 080	LHS 8	1 630
LHSA 8	1 900	LHS10	2 100
LHSA10	2 170	LHS12	2 620
LHSA10M	2 170	LHS14	3 190
LHSA12	2 790	LHS16	3 820
LHSA14	3 540	LHS18	4 610
—	—	LHS20	5 340
—	—	LHS22	6 460

Lubrication

LHSA is prepacked with lubricating grease ALVANIA GREASE 2 (SHELL). LHS is not provided with prepacked grease. Perform proper lubrication for use.
Operating LHS without lubrication will increase the wear of the sliding contact surface or cause seizure.

Operating Temperature Range

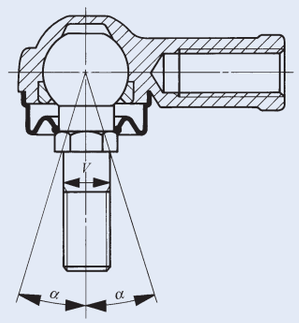
The maximum allowable temperature for L-Balls is +80℃.

Precautions for Use

❶ **Depth of thread**
It is recommended that the depth of thread engagement into the body is more than twice the nominal diameter of thread.

❷ **Permissible angle of tilt**
The permissible angle of tilt is shown in Table 4.

Table 4 Permissible angle of tilt



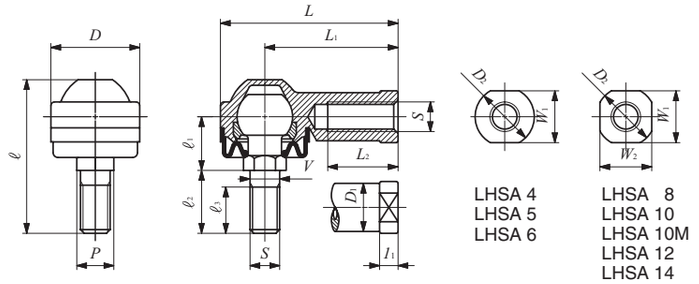
unit: degree

Nominal dia. mm V	LHSA α	LHS α
4	15	—
5	17	15
6	17	17
8	18	18
10	19	19
12	19	19
14	20	20
16	—	20
18	—	21
20	—	20
22	—	21

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

L-BALL





Identification number	Mass (Ref.)	Boundary dimensions mm												
	g	Thread S	V	D	L	L_1	L_2	l_1	W_1	W_2(1)	D_1	D_2	ℓ	P
LHSA 4	11	M 4×0.7	*4	14	25	18	8	4	8	—	8	10	19.5	**5.5
LHSA 5	27	M 5×0.8	5	17	38.5	30	16	5	10	—	10	12	32.5	8
LHSA 6	27	M 6×1	6	19	39.5	30	16	5	10	—	10	12	32.5	8
LHSA 8	64	M 8×1.25	8	24	48	36	19	5	14	14	13	16	41.5	10
LHSA 10	106	M10×1.25	10	28	57	43	23	6.5	17	17	15	19	49	12
LHSA 10M	106	M10×1.5	10	28	57	43	23	6.5	17	17	15	19	49	12
LHSA 12	180	M12×1.75	12	34	67	50	27	6.5	19	19	17.5	22	64	14
LHSA 14	260	M14×2	14	38	76	57	30	8	22	22	20	25	72	17

ℓ_1	ℓ_2	ℓ_3	Ball dia.	Static load capacity C_s N
7	7	5	8	880
12	13	10	11.112	1 180
12	13	10	11.112	1 670
14.5	17	12.5	15	4 380
16	21	17	19.05	7 400
16	21	17	19.05	7 400
20	30	20	22.225	9 900
22.5	33.5	22	25.4	14 600

Note(1) Previous specification does not have the flat surfaces of W_2 dimension.

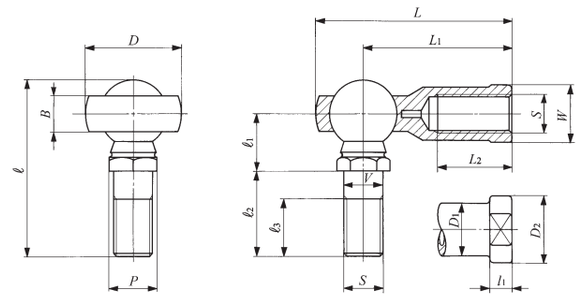
Remarks1. The item marked * is manufactured with a neck diameter of ϕ 3.4. The item marked ** is manufactured with a diameter of ϕ 5.5 instead of a width across flats.

2. Provided with prepacked grease.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

L-BALL





LHS

Identification number	Mass (Ref.)	Boundary dimensions mm												
	g	Thread S	V	D	B	L	L_1	L_2	l_1	W	D_1	D_2	ℓ	P
LHS 5	22	M 5 × 0.8	5	17	6	35.5	27	16	4	9	9	11	30.5	8
LHS 6	32	M 6 × 1	6	19.5	6.75	39.7	30	16	5	11	10	13	36.5	10
LHS 8	60	M 8 × 1.25	8	24	9	48	36	19	5	14	12.5	16	44	11
LHS 10	102	M10 × 1.5	10	28	10.5	57	43	23	6.5	17	15	19	52.5	13
LHS 12	160	M12 × 1.75	12	32	12	66	50	27	6.5	19	17.5	22	61	17
LHS 14	227	M14 × 2	14	36	13.5	75	57	30	8	22	20	25	69	17
LHS 16	300	M16 × 2	16	40	15	84	64	36	8	22	22	27	74	19
LHS 18	445	M18 × 1.5	18	45	16.5	93.5	71	40	10	27	25	31	84	22
LHS 20	580	M20 × 1.5	20	49	18	101.5	77	43	10	30	27.5	34	90.5	24
LHS 22	765	M22 × 1.5	22	54	20	111	84	47	12	32	30	37	99	27

ℓ_1	ℓ_2	ℓ_3	Ball dia.	Static load capacity C_s N
10	15	11	11.112	2 080
11.5	18.5	14	12.7	3 290
14.5	21.5	15	15.875	4 900
17	26	18	19.05	7 640
20	30	20	22.225	12 400
22.5	33.5	22	25.4	14 600
24.5	35.5	23	28.575	19 500
27.5	40.5	25	31.75	25 600
30	43	27	34.925	31 600
32.5	47.5	30	38.1	39 800

Remark Not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch